

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

Roll No. : S018	Name: Khan Shagufta
Class : SYIT	Batch : 1
Date of Assignment : 22/07/2026	Date/Time of Submission : 22/07/2026

Questions:

2a. Write a python code to create nested tuples.

CODE:

```
student=("Shagufta","18",("B.Sc IT", "Semester III"),"73")

print("Name =" , student [0])
print("Roll_No =" , student [1])
print("Department =" , student [2] [0])
print("Semester=" , student [2] [1])
print("Percentage =" , student [3])
```

OUTPUT:

```
Name = Shagufta
Roll_No = 18
Department = B.Sc IT
Semester= Semester III
Percentage = 73
```

2b. Write a python code to sort the nested tuple using sorted () function.

CODE:

```
data =[(1,4),(3,2),(5,1)]
sorted_data = sorted(data,key =lambda x:x[1])
print(sorted_data)
```

OUTPUT:

```
=====
[(5, 1), (3, 2), (1, 4)]
```

2c. Write a python code to copy or clone list.

CODE:

```
File Edit Format Run Options Window Help
list1 = [10,20,30,40]
list2 = list1.copy()
print("original list: ",list1)
print("copied list:",list2)
```

OUTPUT:

```
=====
original list:  [10, 20, 30, 40]
copied list: [10, 20, 30, 40]
```

2d. Write a python code to check immutability property of python tuples.

CODE:

```
File Edit Format Run Options Window Help
fruit = ("Banana", "Apple", "cherry")
fruit[2] = "grapes"
print("fruit")
|
```

OUTPUT:

```
===== RESTART: C:/Users/info/Downloads/Anjalis
Traceback (most recent call last):
  File "C:/Users/info/Downloads/Anjalis/pds_prac1.py", line 2, in <module>
    fruit[2] = "grapes"
TypeError: 'tuple' object does not support item assignment
|
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80